# AQ5.2: Activity Questions 2 - Not Graded

This assignment will not be graded and is only for practice.

**Level 1**

*1 point*

Rank of a matrix $A$A is equal to

- [ ] the row rank of the matrix $A$A.
- [ ] the column rank of the matrix $A$A.
- [ ] the nullity of the matrix $A$A.
- [ ] the number of dependent variables in the system of equations $Ax$ = 0Ax = 0.
- [ ] the number of independent variables in the system of equations $Ax$ = 0Ax = 0.

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

the row rank of the matrix $A$A.

the column rank of the matrix $A$A.

the number of dependent variables in the system of equations $Ax$ = 0Ax = 0.

Let nullity of the matrix $A_{3\times 5}$ A3×5 be 2. Find the rank of the matrix $A_{3\times 5}$ A3×5.

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

(Type: Numeric) 3

*1 point*

Find the rank of the matrix $\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 1 & 4 & 2 \end{bmatrix}$.

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

(Type: Numeric) 2

*1 point*

*1 point*

If $A$A is an $m \times n$m × n matrix, then which of the following statements is true?

- ○ $rank(A) + nullity(A) = m$rank(A) + nullity(A) = m
- ○ $rank(A) + nullity(A) = n$rank(A) + nullity(A) = n
- ○ $rank(A) + nullity(A) = mn$rank(A) + nullity(A) = mn
- ○

$rank(A) + nullity(A) = max\{m, n\}$rank(A) + nullity(A) = max{m, n}

○

$rank(A) + nullity(A) = min\{m, n\}$rank(A) + nullity(A) = min{m, n}

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

$rank(A) + nullity(A) = n$rank(A) + nullity(A) = n

*1 point*

Which of the following is true for a homogeneous system of linear equations $Ax = 0$Ax = 0?

- [ ] It may have no solution.
- [ ] It can have infinitely many solutions.
- [ ] It can have a unique solution.
- [ ] Whenever it has a non-trivial(non zero) solution, it must have infinitely many solutions

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

It can have infinitely many solutions.
It can have a unique solution.
Whenever it has a non-trivial(non zero) solution, it must have infinitely many solutions

*1 point*

Which of the following options are correct for a square matrix $A$A of order $n \times n$n × n, where $n$n is any natural number?

- [ ] If the determinant is non-zero, then the nullity of $A$A must be 0.
- [ ] If the determinant is non-zero, then the nullity of $A$A may be non-zero.
- [ ] If the nullity of $A$A is non-zero, then the determinant of $A$A must be 0.
- [ ] If the nullity of $A$A is non-zero, then the determinant of $A$A may be non-zero.

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

If the determinant is non-zero, then the nullity of $A$A must be 0.
If the nullity of $A$A is non-zero, then the determinant of $A$A must be 0.

*1 point*

Choose the set of correct statements.

- [ ] If nullity of a $3 \times 3$3 × 3 matrix is $c$c for some natural number $c$c, $0 \leq c \leq 3$0 ≤ c ≤ 3, then the nullity of $-A$−A will also be $c$c.
- [ ] $nullity(A + B) = nullity(A) + nullity(B)$nullity(A + B) = nullity(A) + nullity(B)
- [ ] Nullity of the zero matrix of order $n \times n$n × n, is $n$n.
- [ ] Nullity of the zero matrix of order $n \times n$n × n, is 00.
- [ ] There exist square matrices $A$A and $B$B of order $n \times n$n × n, such that nullity of both $A$A and $B$B is 00, but the nullity of $A + B$A + B is $n$n.

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

If nullity of a $3 \times 3$3 × 3 matrix is $c$c for some natural number $c$c, $0 \leq c \leq 3$0 ≤ c ≤ 3, then the nullity of $-A$−A will also be $c$c.
Nullity of the zero matrix of order $n \times n$n × n, is $n$n.
There exist square matrices $A$A and $B$B of order $n \times n$n × n, such that nullity of both $A$A and $B$B is 00, but the nullity of $A + B$A + B is $n$n.
The rank of an $n \times n$n × n matrix all of whose entries are equal to 1 is

**No, the answer is incorrect.**
**Score: 0**
**Accepted Answers:**
(Type: Numeric) 1
*1 point*
*1 point*
If $A$A is a $3 \times 4$3 × 4 matrix, then which of the following options are true?
- [ ] rank(A) must be less than or equal to 3.
- [ ] nullity(A) must be greater than or equal to 1.
- [ ]
If $A$A has 22 columns which are non-zero and not multiples of each other, while the remaining columns are linear combinations of these 22 columns, then $nullity(A)$ = 2nullity(A) = 2.
- [ ]
If $A$A has 22 columns which are non-zero and not multiples of each other, while the remaining columns are linear combinations of these 22 columns, then $nullity(A)$ = 1nullity(A) = 1.

**No, the answer is incorrect.**
**Score: 0**
**Accepted Answers:**
rank(A) must be less than or equal to 3.
nullity(A) must be greater than or equal to 1.
If $A$A has 22 columns which are non-zero and not multiples of each other, while the remaining columns are linear combinations of these 22 columns, then $nullity(A)$ = 2nullity(A) = 2.

**Level 2**

*1 point*
Let $Ax$ = 0Ax = 0 be a homogeneous system of linear equations which has infinitely many solutions, where $A$A is an $m \times n$m × n matrix (where, $m$ > 1, $n$ > 1m > 1, n > 1). Which of the following statements are possible?
- [ ]
$rank(A)$ = $m$rank(A) = m and $m$ < $n$m < n.
- [ ]
$rank(A)$ = $m$rank(A) = m and $m$ > $n$m > n.
- [ ]
$rank(A)$ = $m$rank(A) = m and $m$ = $n$m = n.
- [ ]
$nullity(A)$ = $n$nullity(A) = n.
- [ ]
$nullity(A) \neq 0$nullity(A) = 0.

**No, the answer is incorrect.**
**Score: 0**
**Accepted Answers:**
$rank(A)$ = $m$rank(A) = m and $m$ < $n$m < n.

$nullity(A) = n$nullity(A) = n.
$nullity(A) \neq 0$nullity(A) = 0.

Consider the coefficient matrix $A$A of the following system of linear equations to answer questions 11 and 12:

$3x_1 + 2x_2 + x_3 = 0$ 3x1 + 2x2 + x3 = 0

$x_1 + x_3 = 0$ x1 + x3 = 0

*1 point*

Which one of the following vector spaces represents the null space of $A$A appropriately?

- $\{(-t, t, t) \mid t \in R\}$\{(−t, t, t) | t ∈ R\}
- $\{(t_1, t_2, \frac{t_2 - t_1}{2}) \mid t_1, t_2 \in R\}$\{(t1, t2, 2t2−t1) | t1, t2 ∈ R\}
- $\{(t, -t, t) \mid t \in R\}$\{(t, −t, t) | t ∈ R\}
- $\{(t_1, t_2, \frac{t_1 + t_2}{2}) \mid t_1, t_2 \in R\}$\{(t1, t2, 2t1+t2) | t1, t2 ∈ R\}

**No, the answer is incorrect.**
**Score: 0**
**Accepted Answers:**
$\{(-t, t, t) \mid t \in R\}$\{(−t, t, t) | t ∈ R\}

*1 point*

What will be the rank of $A$A and nullity of $A$A?

- $rank(A) = 3, nullity(A) = 2$rank(A) = 3, nullity(A) = 2
- $rank(A) = 2, nullity(A) = 1$rank(A) = 2, nullity(A) = 1
- $rank(A) = 1, nullity(A) = 2$rank(A) = 1, nullity(A) = 2
- $rank(A) = 2, nullity(A) = 0$rank(A) = 2, nullity(A) = 0

**No, the answer is incorrect.**
**Score: 0**
**Accepted Answers:**
$rank(A) = 2, nullity(A) = 1$rank(A) = 2, nullity(A) = 1

Let $Ax = 0$Ax = 0 be a homogeneous system of linear equations which has a unique solution, where $A \in R^{n \times n}$ A ∈ Rn×n. What is the nullity of $A$A?

**No, the answer is incorrect.**
**Score: 0**
**Accepted Answers:**
(Type: Numeric) 0

*1 point*

Suppose $x_1 \neq 0$x1 = 0 solves $Ax = 0$Ax = 0, where $A \in R^{2 \times 4}$ A ∈ R2×4. What is the minimum number of elements in a linearly independent subset of null space of $A$A that also spans the set of solutions of $Ax = 0$Ax = 0?

**No, the answer is incorrect.**
**Score: 0**
**Accepted Answers:**
(Type: Numeric) 2
*1 point*
*1 point*
Consider the following sets:
$S_1 = \{(1,2,3),(4,5,6),(8,9,10)\}$S1 $= \{(1,2,3),(4,5,6),(8,9,10)\}$
$S_2 = \{(1,2,1),(2,4,2),(1,2,5)\}$S2 $= \{(1,2,1),(2,4,2),(1,2,5)\}$
$S_3 = \{(1,2,3),(4,5,6),(7,8,10)\}$S3 $= \{(1,2,3),(4,5,6),(7,8,10)\}$
$S_4 = \{(1,2),(4,5)\}$S4 $= \{(1,2),(4,5)\}$
$S_5 = \{(1,2),(4,5),(3,8)\}$S5 $= \{(1,2),(4,5),(3,8)\}$
Choose the set of correct options

- [ ] $S_1$ S1 is a basis of $R^3$ R3.
- [ ] $S_2$ S2 is a basis of $R^3$ R3.
- [ ] $S_3$ S3 is a basis of $R^3$ R3.
- [ ] $S_4$ S4 is a basis of $R^2$ R2.
- [ ] $S_5$ S5 is a basis of $R^2$ R2.

**No, the answer is incorrect.**
**Score: 0**
**Accepted Answers:**
$S_3$ S3 is a basis of $R^3$ R3.
$S_4$ S4 is a basis of $R^2$ R2.

Check Answers